



ONLINE COURSE:

**APPLICATION OF GIS
IN CONSERVATION
MAPPING**

**“Towards the Achievement of
Target 3 of the Global Biodiversity
Framework.”**

The banner features a background image of a gazelle on the left and a map of Eastern and Southern Africa on the right. The map shows various countries and water bodies, with a red line indicating a specific path or boundary. A small inset image of a cheetah is visible in the top right corner of the map area.

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GIS Biodiversity Ecosystems
Expert



Module 1

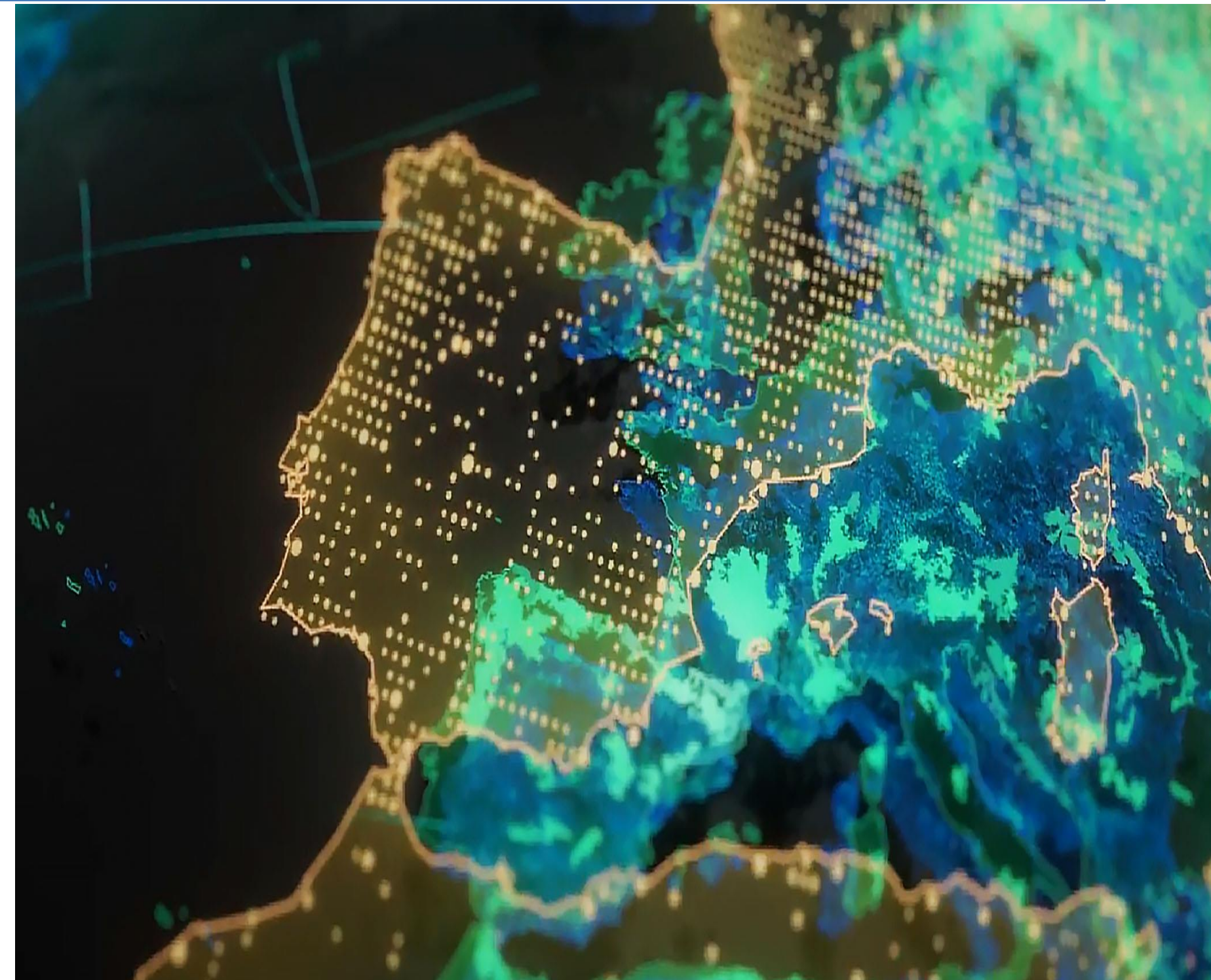
Conservation Challenges & Global Biodiversity Framework

Session 3: GIS & RS

Introduction to GIS and Remote Sensing

Definition: GIS

- ❑ GIS (Geographic Information Systems) is a system designed to capture, store, manipulate, analyze, manage, and present spatial or geographic data.
- ❑ Integrates various data types
- ❑ Allows visualization through maps and spatial analysis.



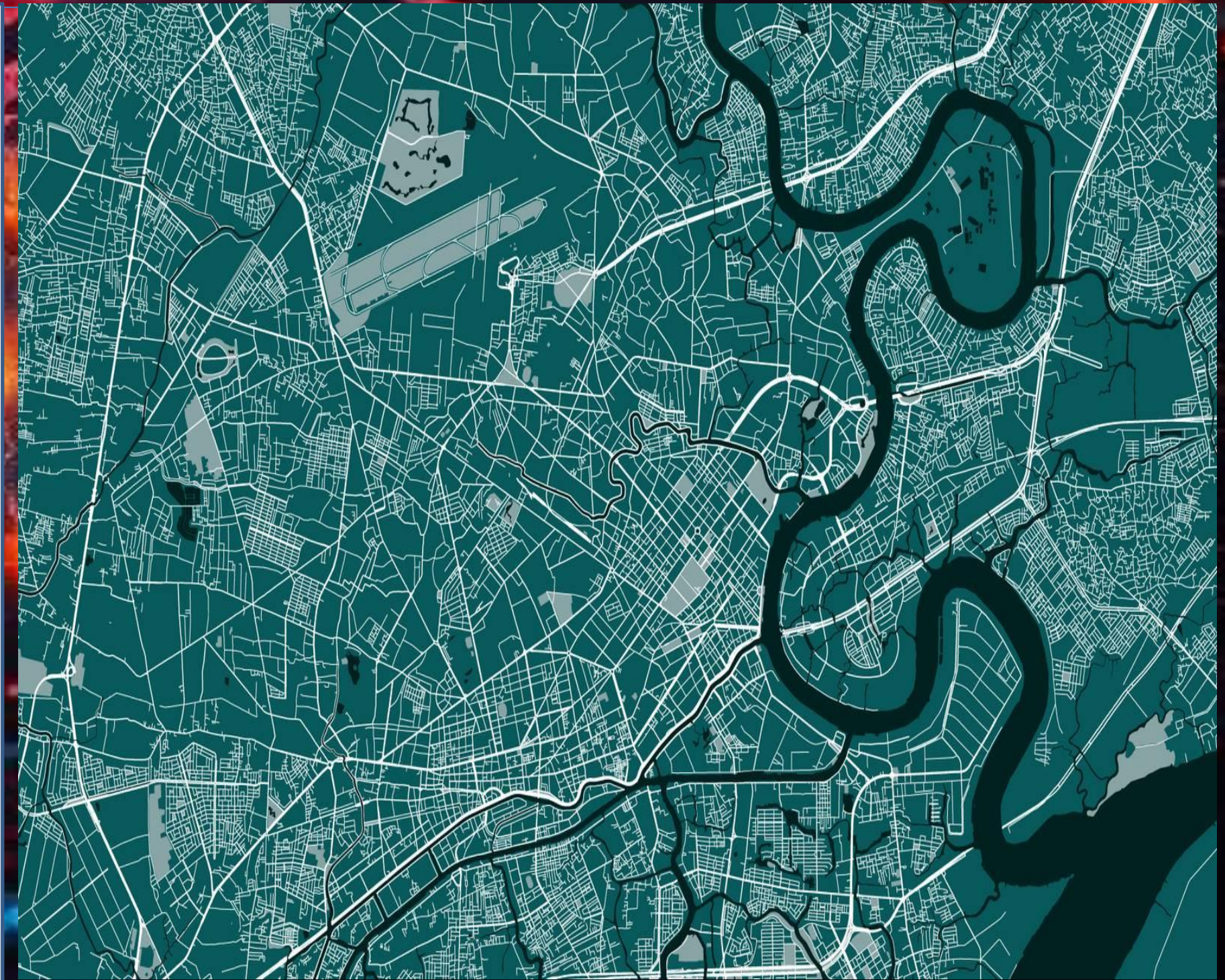
Definition: Remote Sensing

- ❑ Remote sensing is the process of detecting and monitoring the physical characteristics of an area by measuring its reflected and emitted radiation from a distance.
- ❑ Used to monitor land use, vegetation, climate, and changes over time.



Benefits

- ❑ Maximizing the efficiency of planning and decision making
- ❑ Integrating information from multiple sources
- ❑ Facilitating complex querying and analysis
- ❑ Eliminating redundant data and minimizing duplication



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Components of GIS

❑ GIS has five components:

- ❑ - Hardware
- ❑ - Software
- ❑ - Data
- ❑ - People
- ❑ - Procedures



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Components of GIS: Hardware

❑ It consists of the equipment and support devices that are required to capture, store process and visualize the geographic information. These include computer with hard disk, digitizers, scanners, printers and plotters etc.



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Components of GIS: Software

□ These are programs that handle spatial data. The GIS software must have the basic capabilities of data input, storage, transformation, analysis and provide desired outputs. The interfaces could be different for different software's.



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Components of GIS: Software

❑ These are programs that handle spatial data. The GIS software must have the basic capabilities of data input, storage, transformation, analysis and provide desired outputs. The interfaces could be different for different software's.

❑ Types of Software:

- ❑ Commercial / Proprietary Software
- ❑ Open Source Software



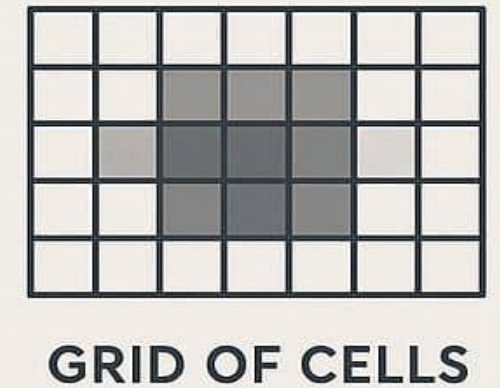
- ❑ Spatial data refers to facts and figures that contain location information.
- ❑ Location refers to position of objects or processes on the earth surface. Types of location: Absolute and Relative location.
- ❑ Elements of Spatial data: Geometry, Attributes, and Metadata.
- ❑ GIS data models: Vector and Raster

GIS DATA TYPES

VECTOR



RASTER



ATTRIBUTE

NAME	VALUE	2
_____	_____	—
_____	_____	—
_____	_____	—

METADATA

- SOURCE
- FORMAT
- DESCRIPTION

Components of GIS: People

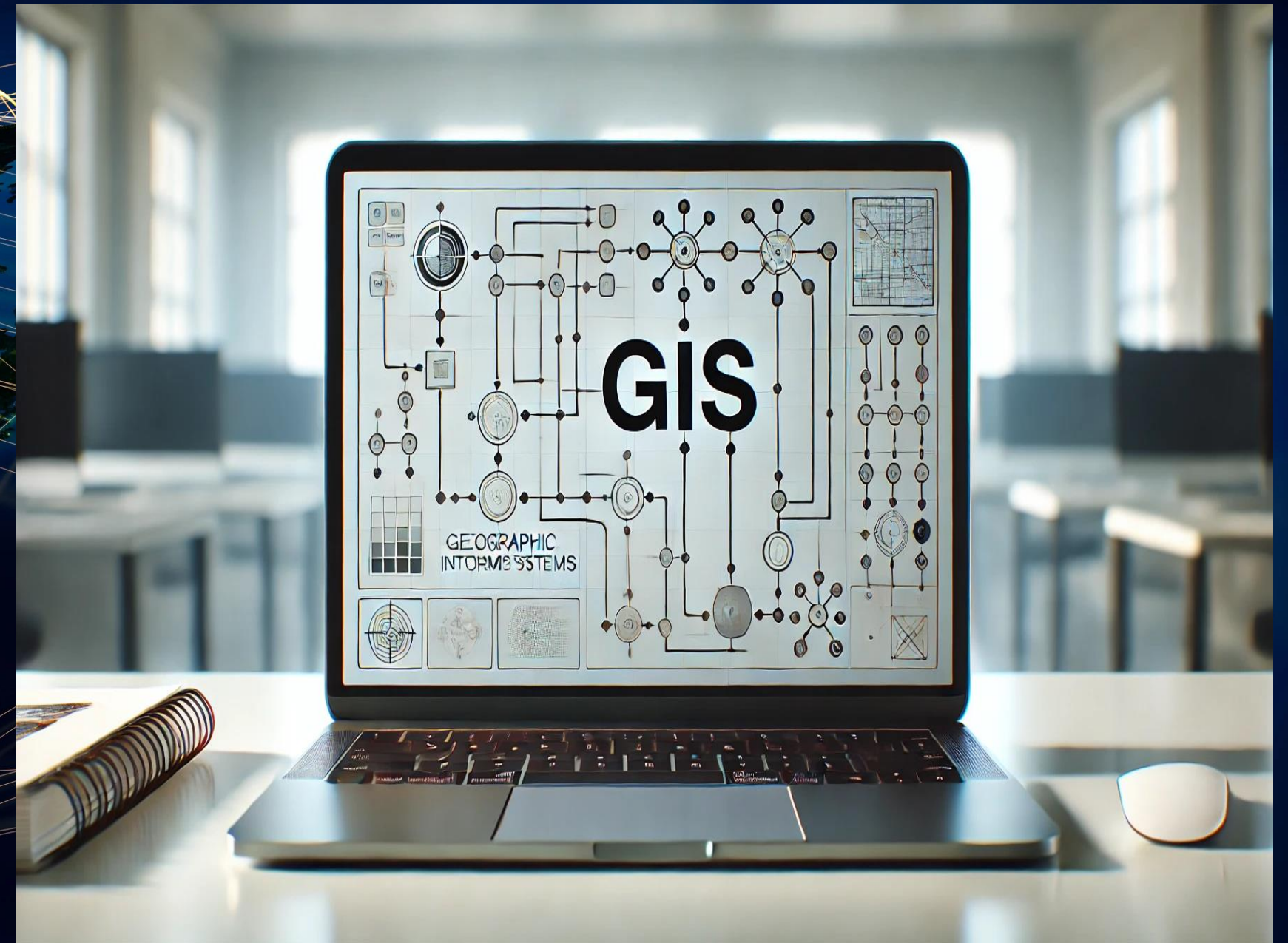
- ❑ People refers to the users and developers of the geographic Information System.
- ❑ Developers can be categorized as hardware or Software developers
- ❑ Users can be categorized in terms of expertise or specialization



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Procedures and Policies

❑ These are methods and workflows used in Geographic Information Systems (GIS) for managing, analyzing, and visualizing spatial data



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Functions of GIS

- ❑ Functions of GIS can be summarized into 4:
 - ❑ Data capture and preparation
 - ❑ Data storage and management
 - ❑ Data analysis and querying
 - ❑ Data presentation



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Data Capture

❑ Entails:

- ❑ Processes & methods of data acquisition such as surveying, photogrammetry, remote sensing, and digitizing
- ❑ Remote sensing and photogrammetry provide images and photographs as the raw base data from which spatial data sets are derived.
- ❑ Surveying is employed to collect data that cannot be obtained with remote sensing techniques e.g. socio-economic data or very precise data, or to validate data obtained through RS and photogrammetry.
- ❑ Traditional techniques for obtaining spatial data, typically from paper sources, included manual digitizing and scanning.



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Data storage and management

❑ In a GIS, features are represented with their geometric and non-geometric attributes and relationships. This representation has to be considered when storing geographic data

❑ In most of the available systems, spatial data is organized in layers by theme and/or scale. For instance, the data may be organized in thematic categories, such as land use, topography and administrative subdivisions, or according to map scale.

**Spatial Data
Management**



**Spatial
Databases**

*Manage GIS Data
Efficiently*

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Data analysis and Manipulation

- ❑ Analysis of spatial data can be defined as computing new information that provides new insight from the existing, stored spatial data.
- ❑ The analysis functions of a GIS use the spatial and non-spatial attributes of the data in a spatial database to provide answers to user questions.

**Spatial Data
Management**



**Spatial
Databases**

*Manage GIS Data
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COORDINATE SYSTEMS AND MAP PROJECTIONS

❑ Map Datum

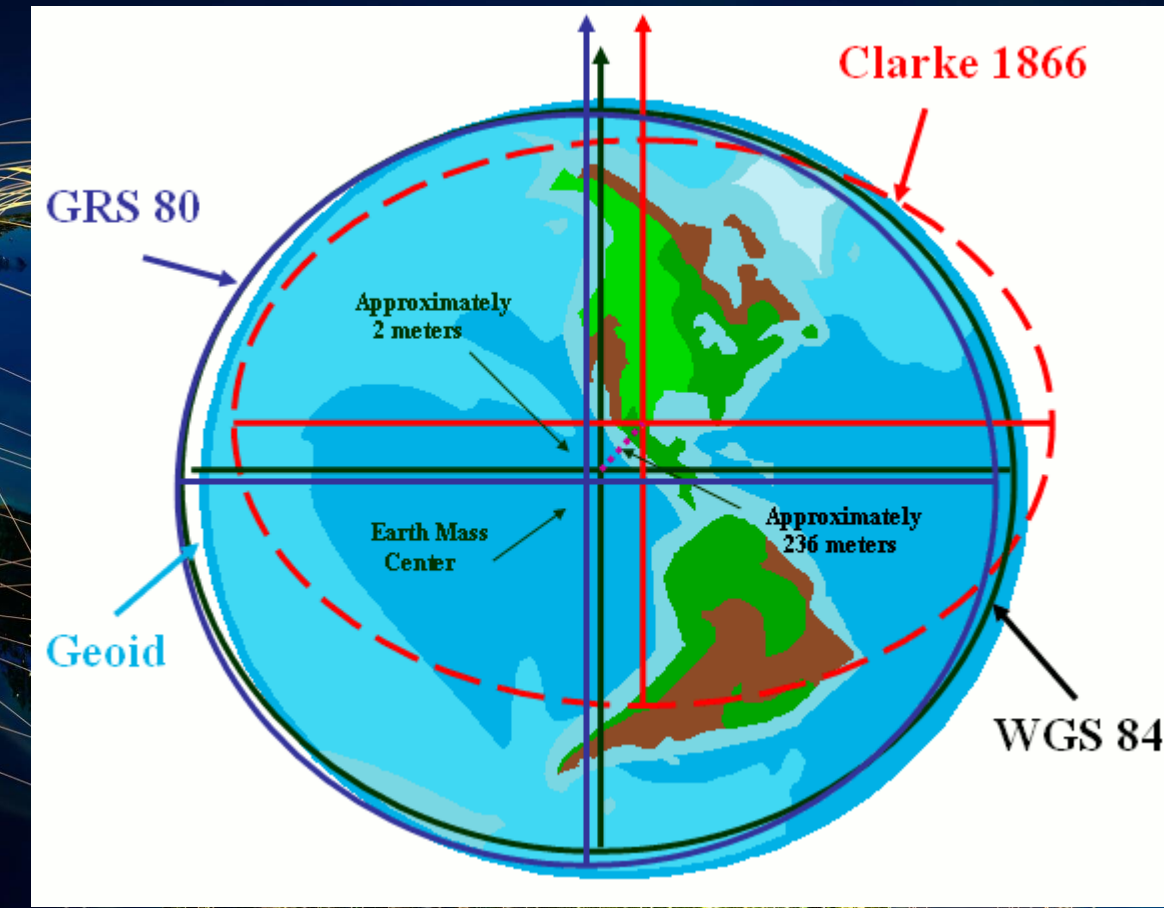
❑ A datum is a reference point or surface against which measurements are made using models of the shape of the earth.

❑ a) Vertical

❑ B) Horizontal

❑ Local geodetic datum

❑ Global/Geocentric datum



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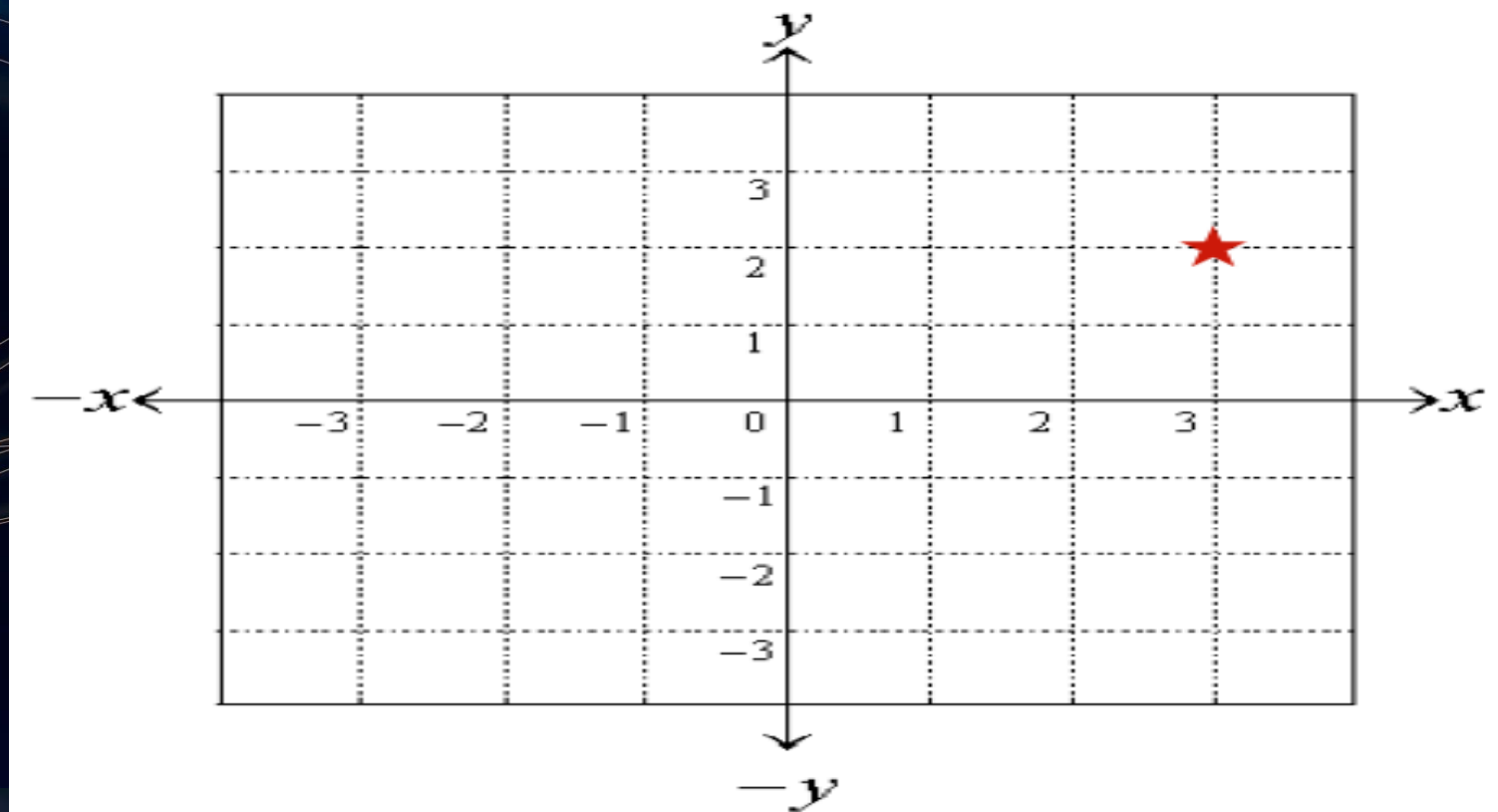
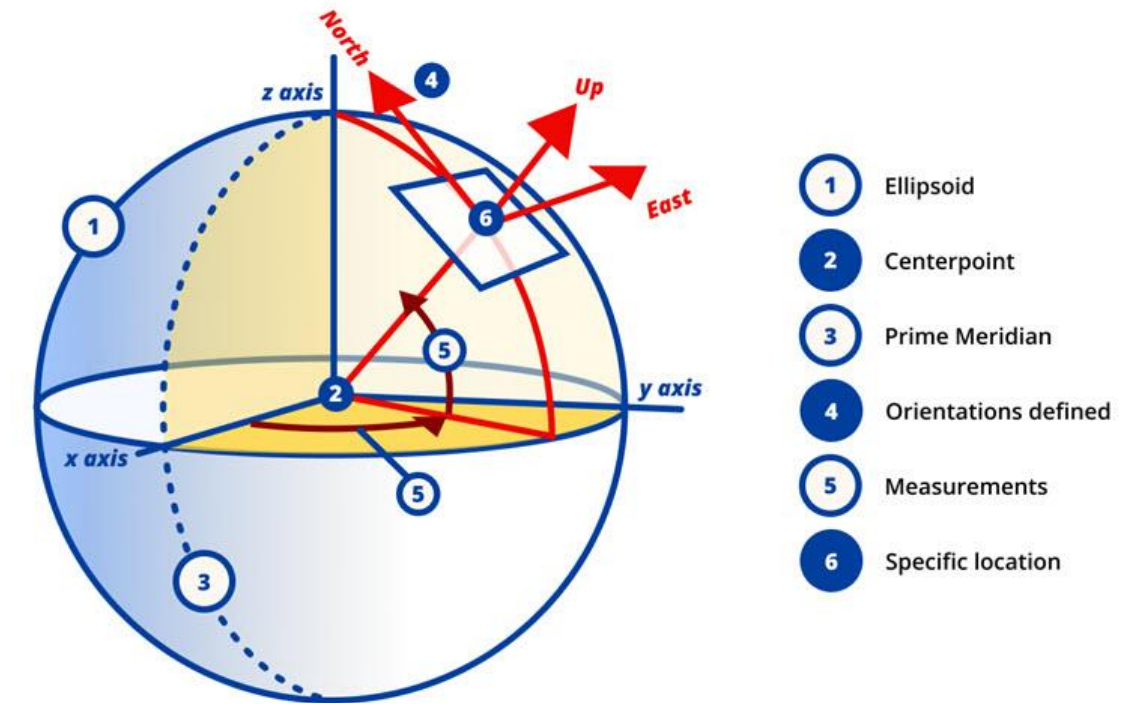
COORDINATE SYSTEMS AND MAP PROJECTIONS

❑ Coordinate System

❑ A coordinate system is a reference system used for locating objects in a two or three-dimensional space

❑ A geographic coordinate system, also known as global or spherical coordinate system is a reference system that uses a three-dimensional spherical surface to determine locations on the earth.

❑ A projected coordinate system is defined as two-dimensional representation of the Earth. It is based on a spheroid geographic coordinate system, but it uses linear units of measure for coordinates.



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